

OSLT Research Engine v2.3 RC1

Executive summary: multidisciplinary reasoning architecture, anti-drift controls and required evidence estate

640	100	15	65	13	16
ontology variables	analytical methods	domain kernels	source families	acquisition workstreams	certainty dimensions

Purpose. OSLT is a governed scientific operating architecture for complex, multicausal research. It separates evidence acquisition, causal identification, historical and linguistic interpretation, adversarial challenge, certainty calibration and release. It is research-only: it is not a diagnostic or clinical decision system.

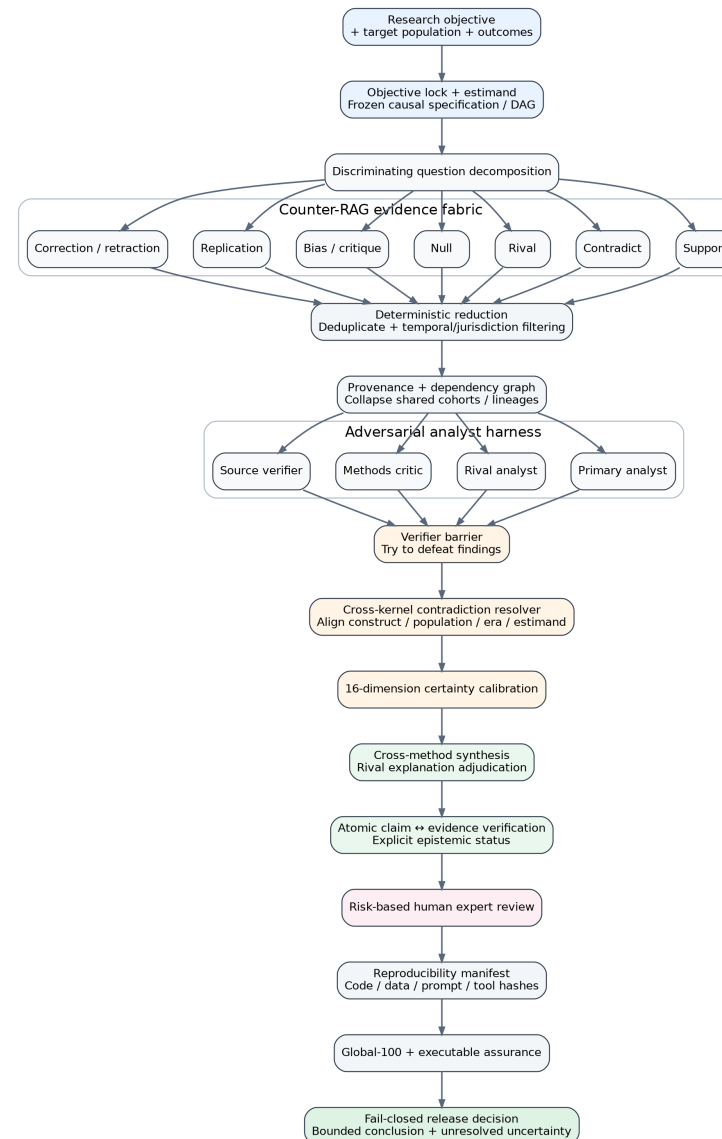
Central rule. A proposition is not strengthened merely because more papers, more analyses or more AI agents agree. OSLT asks whether evidence discriminates the proposition from its strongest rival explanation, whether the evidence is independent, and what maximum claim the design permits.

Engineering validation	Result
Unit/regression tests	171 / 171 passing
Exact product-code coverage	95.20%
Targeted mutation campaign	12 / 12 detected
Global-100 architecture audit	100 / 100
Executable assurance overlay	20 / 20

These are engineering-control results, not empirical validation of any substantive hypothesis.

1. How the engine works

The workflow is a graph, not a single long prompt. Independent retrieval and analyst tasks fan out; deterministic code performs plumbing; verifier and release gates sit on the edges before claims can move downstream.



2. Reasoning safeguards and anti-drift loop

Evidence before conclusion

Seven Counter-RAG lanes independently retrieve support, contradiction, rivals, null findings, criticism, replication and corrections.

Dependency-aware synthesis

Shared cohorts, datasets, research groups and guideline ancestry are collapsed before triangulation.

Context-aware contradiction

Apparent disagreements are first checked for construct, population, time, jurisdiction and estimand mismatch.

Bounded interpretation

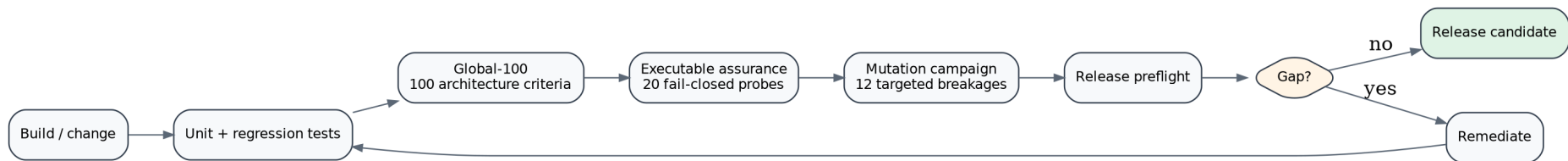
Historical, discourse, narrative and psychodynamic methods can contribute evidence and hypotheses but cannot self-authorise population causal claims.

Certainty is multidimensional

Sixteen certainty dimensions remain separate; OSLT does not compress causal certainty, measurement quality and explanatory contribution into one opaque percentage.

Fail-closed release

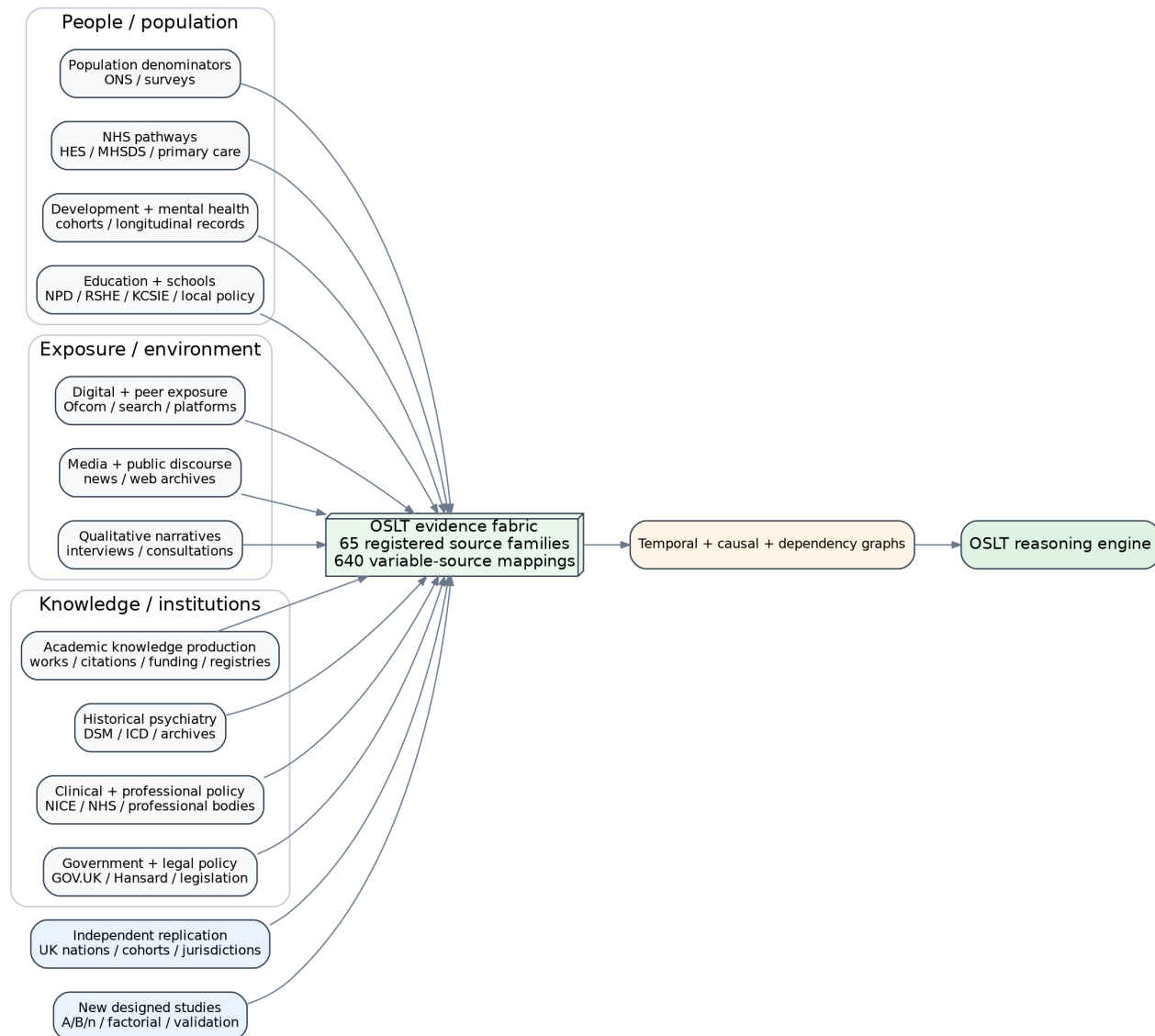
High-impact findings can require human review; reproducibility, claim verification and anti-drift gates must pass before release.



Global-100 remains an architecture anti-drift audit. v2.3 adds a 20-probe executable assurance overlay to check selected load-bearing controls actually fail closed.

3. What data must be accumulated

OSLT requires a heterogeneous evidence estate because the central research question may have interacting population, clinical, developmental, institutional, educational, digital, historical and epistemic mechanisms.



4. Data acquisition workstreams

ID	Workstream	Data to accumulate	Priority
W01	Population denominators and trend decomposition	Population size, age, sex, geography, identity/prevalence measures, survey denominators, time trends	P0
W02	NHS referrals, diagnoses and service pathways	Referral dates, service contacts, diagnoses, procedures, waiting times, provider/geography, outcomes	P0
W03	Primary care and psychiatric/neurodevelopmental history	GP records, psychiatric diagnoses, autism/ADHD, medication, symptoms, trauma proxies, prior service use	P0
W04	Longitudinal development, family and cohort data	Puberty/development, mental health, family, peers, school, social context, repeated measures, genetics where justified	P0/P1
W05	Education, safeguarding and school implementation	NPD outcomes, school policy, RSHE materials, safeguarding guidance, implementation dates, local variation	P0
W06	Digital, social-media and peer exposure	Platform use, search behaviour, recommendation exposure where obtainable, peer-network measures, media-use surveys	P0/P1
W07	Academic knowledge-production and publication selection	Works, authors, institutions, citations, funding, preregistration, trials, reviews, retractions, publication timing	P0/P1
W08	Historical psychiatry, diagnostic nosology and treatment doctrine	DSM/ICD definitions, textbooks/archives, diagnostic criteria, differential diagnosis, treatment recommendations, semantic change	P0/P1
W09	Clinical guidelines, professional bodies and NHS policy	NICE, NHS specifications, Cass implementation, professional guidance, commissioning/service documents	P0
W10	Government, legal and parliamentary policy corpus	GOV.UK consultations, archived government pages, Hansard, legislation, FOI material	P0/P1
W11	Media and public discourse	News corpora, web archives, GDELT, lexical/semantic/framing data over time	P1
W12	Primary qualitative and narrative evidence	Longitudinal interviews, clinical/patient narratives where lawfully available, clinician accounts, coded consultation material	P0
W13	Designed experiments and replication studies	Preregistered vignettes, framing A/B/n, factorial designs, measurement validation, independent replication	P0

5. Discovery sequence and next phase

Phase	Action	Purpose
1	Freeze definitions	Define populations, outcomes, constructs, time zero, estimands, rival hypotheses and versioned causal assumptions.
2	Acquire open evidence	Population aggregates, literature metadata/full text, policy/guidelines, historical corpora and public media/digital sources.
3	Build graphs	Construct temporal, semantic, citation/dependency, institutional and causal graphs before synthesis.
4	Obtain restricted microdata	Secure NHS/ONS/DfE/cohort access and verify field-level variables, linkage and governance.
5	Decompose the observed trend	Separate prevalence, population composition, ascertainment, referral/service and coding effects before attributing mechanisms.
6	Run blinded domain analyses	Use the 15 domain kernels and routed analytical methods with Counter-RAG and adversarial verification.
7	Replicate and stress-test	Use independent cohorts/jurisdictions, multiverse/specification checks, negative controls and transportability tests.
8	Commission missing discriminatory evidence	Use preregistered A/B/n, factorial, measurement or qualitative studies only for unresolved mechanism questions.
9	Synthesize cautiously	Resolve contradictions, calibrate certainty, verify atomic claims and preserve an explicit unexplained component.

Readiness judgement

The conceptual/engineering architecture is now sufficiently developed that further gains are more likely to come from empirical qualification than from adding more abstract layers. The next critical work is to build gold-standard model/retrieval benchmarks, secure the P0 data sources, verify field-level constructs, and run preregistered pilot studies through the complete graph.

Important limitation: OSLT v2.3 has not itself established why any observed population trend occurred. It is an architecture for conducting that investigation under explicit methodological and evidential controls.